

EDUCATION

- Carnegie Mellon University, Pittsburgh, PA Class of 2030
 - Bachelors of Sciences – Intended majors in Physics and Robotics
- The Harker School, San Jose, CA Class of 2026
 - Relevant Coursework: Neural Networks, Nanoscience, Differential Equations, Multivariable Calculus, AP Physics, Data Structures, Honors Biology, Honors Chemistry

SKILLS

- Software: Mathematica, Fusion 360 CAD
- AI: Hermes, OpenClaw, Nebius, HydraDB, Slate
- Lab: Chemical etching, microscopy and imaging

RESEARCH AND LAB EXPERIENCE

- **Student Researcher @ SLAC National Accelerator Laboratory** **July 2026 - August 2026**
Neilander Group
 - Fabricated anti-reflective Si surfaces for solar cells by performing anisotropic 2M KOH wet etching on single-crystal Si wafers, then imaged the resulting pyramidal surface texture via scanning electron microscopy (SEM).
 - Built TiO₂/p-type silicon heterojunction solar cells from etched and unetched Si wafers and measured open-circuit voltage for each.
 - Calculated the theoretical optical benefit of the etched texture, showing a promising reflectance reduction from about 30% to about 9%.
 - Ran electrolysis experiments on zinc, tin, aluminum, and copper electrodes in an electrolyte solution, observing and documenting corrosion and degradation behavior.
- **Materials Science Researcher and Head of Microscopy** **May 2025 - May 2026**
Starostina Lab, Aspiring Scholars Directed Research Program
 - Coauthored “Chemical Etchants for Dihedral Angle Measurements of Twin Boundaries in Pure Copper,” a copper etchants paper accepted to IEEE MIT Research Technology Conference.
 - Presented findings to 300+ people at research expositions, conferences, and colloquia.
 - Created and taught a 24-student course on microscopy, optical interferometry, and CAD.
 - Tested multiple copper etchants including HNO₃, FeCl₃, and HCl solutions with copper and zinc.
- **Optics Researcher @ University of Cambridge** **October 2024 - September 2025**
Alban-Paccha Group
 - Conducted a literature review comparing three optical interconnect modulator technologies: VCSELs, MOS-capacitor, and plasmonic Mach-Zehnder, on data rate, bandwidth, energy-per-bit, and voltage-length product.
 - Selected as one of eight high school students to attend the 2025 Chinese American Semiconductor Professional Association Banquet for this research.
- **Biophysics Summer Researcher @ Keck Graduate Institute** **June 2024 - July 2024**
Niemz Lab and Casper Lab
 - Designed an NGS based screening procedure for non-invasive prenatal detection of congenital heart defects.
 - Authored an industry style cost and risk evaluation for a drug repurposing proposal (repurposing disulfiram for Lyme disease).
 - Presented findings to two committees of KGI investors and professors.
- **Personal Engineering Invention** **September 2022 - November 2023**
Friction reducing hairbrush
 - Engineered a friction reducing hairbrush that forms air bearings around each bristle.
 - Built multiple prototypes, engineering custom circuits and mechanisms to tune its performance.
 - Applied for a patent now pending 2nd Office Action.

LEADERSHIP

- **Founder, Discovery Day** **February 2025 - Present**
[bay-area-discovery-day.vercel.app](#)
 - Founded an annual day of hands-on chemistry and physics workshops at Peterson Middle School to counter science department budget cuts, growing the program to 6 directors and reaching 500+ students thus far (30% English second language).
- **Founder, Community Tables Cooking Classes** **August 2024 - April 2026**
[community-tables.vercel.app](#)
 - Founded and taught monthly bilingual (Spanish and English) hands-on cooking classes for 300+ families at Sunday Friends, a food distribution and family services center in East San Jose, raising \$800 in grant funding from NPO sponsors.
- **Board Member, National Charity League** **April 2021- April 2023**
President (April 2023 - 2025), VP of Philanthropy (April 2021 - 2023)
 - Led a 140+ member mother daughter chapter, coordinating 800+ service hours across shelters, food banks, and education programs.
 - Raised \$27,000 over 5 years for Walk to End Alzheimer's thus far through community events.

HONORS AND AWARDS

- 1st Prize Winner - Bay Builders Hackathon, AWS Builders Center - July 2026
- STEM Merit Scholarship (5% of Applicants)
 - by University of Cambridge, for a year-long optical physics research project
- Presidential Volunteer Service Award (National): Gold - 2022, 2023, 2024
- \$800 Inspire Innovation Grant Recipient - August 2024
 - by National Charity League, to one applicant whose initiative is impactful and transformative